

Figure 1: Block-level architecture—MSC device to Linux host

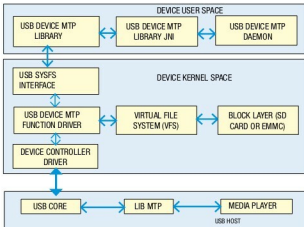


Figure 2: Block-level architecture—the MTP device to host

in Media Transfer Protocol does not just mean audio/video; it encompasses all binary data, including text files. Devices supporting MTP come up as a ‘portable device’, and not a drive as in USB MSC. All the content (songs, images, videos) on devices are represented by objects, and object handles are used in order to reference a logical object on the device. Once the device is connected to the USB host, enumeration takes place, and all associated metadata, such as the file creation time, the time of modification, size, folder details, etc. are passed to the host PC. After this, data exchange can begin.

Android implementation of MTP

In Linux, sysfs is an interface used for communication between user space and kernel space. Figure 2 illustrates the block-level architecture when an MTP device is connected to a standard host. The MTP implementation on the device side is divided across user space and kernel space.

User space components: The *mtpd* daemon (*mtpd*) runs in the JVM (Java Virtual Machine) of Android. It loads and calls the USB Device MTP Library via JNI (Java Native Interface). The USB Device MTP Library takes care of decoding and responding to MTP commands.

Kernel space components: The USB Device MTP Function Driver handles the MTP class-specific control requests like opening or closing the session as per the user’s instructions. It also controls the reading/writing of files from the storage media through the Virtual File System.

On the host side, the USB core drivers communicate with the USB device. Applications like Windows Media Player on Windows use the *libmtp* library to communicate with the device. Based on the requests from the applications, *mtpd* on the device would respond using the USB drivers.

The advantages of MTP

From both the architectures, you can see that MSC works on SCSI commands, and the host has unrestricted access to the device. The device cannot modify the data without releasing its connection from the host, if mounted. If the device is given permissions to modify the data when there is a connection with the host, there are chances of corrupting the data, or even the file system. That is why, when you mount an SD card of an Android device to a PC, you cannot access the SD card using the Android device until you unmount it from the PC. MTP has a solution to this problem. It manages the files, unlike MSC, which manages storage. The media player on the host reads and writes entire files on the device.

With MTP, Windows Media Player has an option to sync media content on the PC and the device. You can even customise the data content for the sync options based on the ratings, etc. With MSC, this can be achieved only with special applications.

Unlike MSC, MTP enables the monitoring of device-initiated events, and the changes in device properties.

MTP does not allow the transfer of unsupported file formats, unlike mass storage, where the user needs to check for compatibility after the file transfer.

Lastly, MSC devices are more prone to virus attacks than MTP.

USB MSC and MTP represent two efficient data exchange methods. While MSC has been the de-facto standard followed until the recent past, it fails to address the needs posed by modern-day intelligent storage devices. While flash-based devices will continue to use MSC, we will see more and more portable devices with hard drives switching over to MTP. However, it is always the user’s prerogative to choose between the free way or the safe way to manage data, i.e., whether to use MSC or MTP. 

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